

## 1. Core Requirements

A one-dimensional abrupt silicon p-n homojunction was modeled at ( $T = 300$  K). The p-side and n-side doping concentrations were chosen as

$$N_A = 1.0 \times 10^{17} \text{ cm}^{-3}$$

$$N_D = 1.0 \times 10^{16} \text{ cm}^{-3}$$

The depletion approximation was used for the equilibrium electrostatic calculation.

The p-type semiconductor occupies ( $x < 0$ ), the n-type semiconductor occupies ( $x > 0$ ), and the metallurgical junction is located at ( $x = 0$ ).

### Parameters Used in the Simulation

| Parameter                       | Symbol                   | Value                                |
|---------------------------------|--------------------------|--------------------------------------|
| Temperature                     | $T$                      | 300 K                                |
| Elementary charge               | $q$                      | $1.602 \times 10^{-19} \text{ C}$    |
| Boltzmann constant              | $k_B$                    | $1.381 \times 10^{-23} \text{ J/K}$  |
| Silicon relative permittivity   | $\epsilon_{r,\text{Si}}$ | 11.7                                 |
| Silicon band gap                | $E_g$                    | 1.12 eV                              |
| Intrinsic carrier concentration | $n_i$                    | $1.0 \times 10^{10} \text{ cm}^{-3}$ |
| Acceptor concentration          | $N_A$                    | $1.0 \times 10^{17} \text{ cm}^{-3}$ |
| Donor concentration             | $N_D$                    | $1.0 \times 10^{16} \text{ cm}^{-3}$ |
| Junction area                   | $A$                      | $1.0 \times 10^{-4} \text{ cm}^2$    |
| Electron diffusion coefficient  | $D_n$                    | $35 \text{ cm}^2/\text{s}$           |
| Hole diffusion coefficient      | $D_p$                    | $12 \text{ cm}^2/\text{s}$           |
| Electron diffusion length       | $L_n$                    | 10 $\mu\text{m}$                     |
| Hole diffusion length           | $L_p$                    | 10 $\mu\text{m}$                     |

| Extension parameter         | Symbol              | Value                                  |
|-----------------------------|---------------------|----------------------------------------|
| SRH lifetime                | $\tau_{\text{SRH}}$ | $1 \times 10^{-6} \text{ s}$           |
| Series resistance           | $R_s$               | $5.0 \Omega$                           |
| Breakdown voltage           | $V_{BR}$            | 20V                                    |
| Breakdown current scale     | $I_{BR0}$           | $1 \times 10^{-6} \text{ A}$           |
| Breakdown sharpness voltage | $V_A$               | 1 V                                    |
| Richardson constant         | $A^{**}$            | $112 \text{ A cm}^{-2} \text{ K}^{-2}$ |
| Schottky barrier height     | $\phi_{BN}$         | 0.55eV                                 |

|                             |          |     |
|-----------------------------|----------|-----|
| M–S contact ideality factor | $n_{MS}$ | 1.1 |
|-----------------------------|----------|-----|

### (a) Equilibrium Electrostatics

The thermal voltage at (300 K) is

$$V_T = \frac{k_B T}{q} = 0.02585 \text{ V.}$$

The built-in potential of an abrupt p-n junction is calculated from

$$V_{bi} = V_T \ln \left( \frac{N_A N_D}{n_i^2} \right)$$

Using the parameters listed above, the calculated built-in potential is

$$V_{bi} = 0.77384 \text{ V.}$$

Under the depletion approximation, the total depletion width is

$$W = \sqrt{\frac{2\epsilon_s}{q} \left( \frac{1}{N_A} + \frac{1}{N_D} \right) V_{bi}}$$

The depletion widths on the p-side and n-side are

$$x_p = \frac{N_D}{N_A + N_D} W, \quad x_n = \frac{N_A}{N_A + N_D} W.$$

The calculated values are

$$x_p = 0.03016 \text{ } \mu\text{m},$$

$$x_n = 0.30162 \text{ } \mu\text{m},$$

$$W = x_p + x_n = 0.33178 \text{ } \mu\text{m}.$$

The charge density is described by

$$\rho(x) = -qN_A, \text{ \& } -x_p \leq x < 0, \text{ \& } +qN_D, \text{ \& } 0 \leq x \leq x_n$$

*0, \& outside the depletion region*

The electric field is obtained from Poisson's equation,

$$\frac{dE}{dx} = \frac{\rho(x)}{\epsilon_s},$$

and the electrostatic potential satisfies

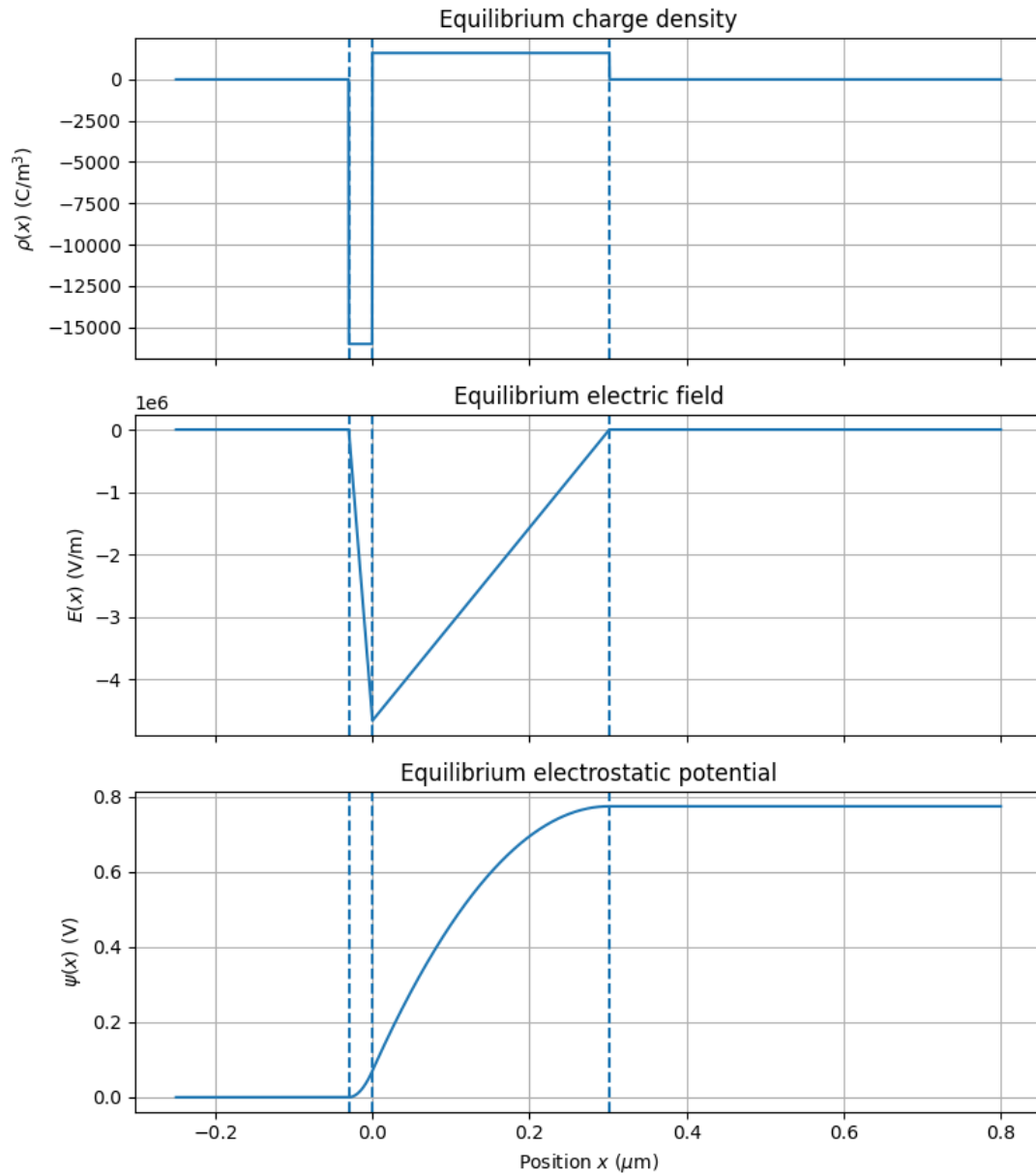
$$E(x) = -\frac{d\psi}{dx}.$$

The maximum electric-field magnitude occurs at the metallurgical junction and is

$$E_{\max} = \frac{qN_A x_p}{\epsilon_s} = \frac{qN_D x_n}{\epsilon_s}.$$

The simulated maximum electric field is

$$E_{\max} = 4.6648 \times 10^4 \text{ V/cm}.$$



**Figure 1.** Simulated equilibrium charge density  $\rho(x)$ , electric field  $E(x)$ , and electrostatic potential  $\psi(x)$  of the abrupt silicon p-n junction.

The p-side depletion region contains negatively charged ionized acceptors, whereas

the n-side depletion region contains positively charged ionized donors. Since  $N_A$  is ten times larger than  $N_D$ , the depletion region extends mainly into the more lightly doped n-side. The simulated result satisfies the charge-neutrality condition

$$N_A x_p = N_D x_n.$$

Therefore,

$$\frac{x_n}{x_p} = \frac{N_A}{N_D} = 10,$$

which agrees with the simulated depletion widths.

The electric field is negative because it points from the positively charged n-side toward the negatively charged p-side. It is zero at both depletion edges and reaches its maximum magnitude near ( $x=0$ ). The electrostatic potential increases from the p-side to the n-side by approximately  $V_{bi}$

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## (b) Equilibrium Band Diagram

The energy-band edges are related to the electrostatic potential by

$$E_C(x) = -q\psi(x) + C,$$

where (C) is an arbitrary energy reference. When energy is expressed in electron volts, the spatial variation can be written as

$$E_C(x) = -\psi(x) + C.$$

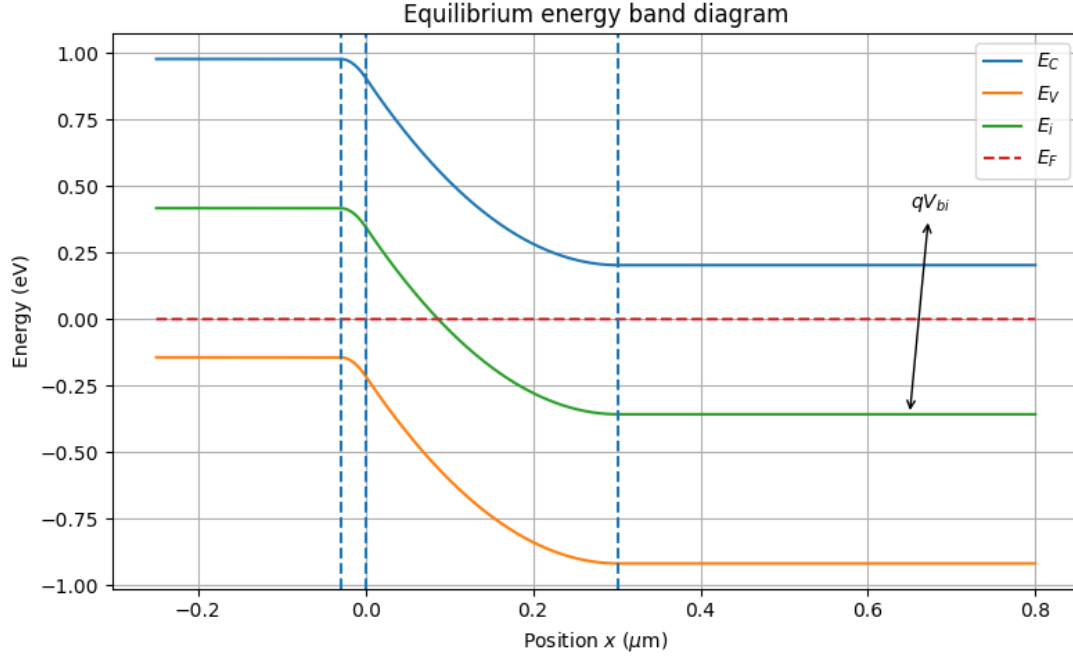
The valence-band edge and intrinsic energy level were calculated using

$$E_V(x) = E_C(x) - E_g,$$

$$E_i(x) = \frac{E_C(x) + E_V(x)}{2}.$$

At thermal equilibrium, the Fermi level is constant throughout the device:

$$E_F(x) = \text{constant}.$$



**Figure 2.** Equilibrium energy-band diagram showing the conduction-band edge  $E_C$ , valence-band edge  $E_V$ , intrinsic energy level  $E_i$ , and Fermi level  $E_F$ .

The conduction-band edge, valence-band edge, and intrinsic energy level bend downward from the p-side to the n-side. The total band bending is  $qV_{bi} = 0.77384$  eV.

The Fermi level remains flat because the device is in thermal equilibrium and there is no net current flow. On the p-side,  $E_F$  is closer to the valence-band edge, while on the n-side,  $E_F$  is closer to the conduction-band edge. This behavior is consistent with the p-type and n-type doping conditions.

The energy bands bend only inside the depletion region because the electric field is approximately zero in the quasi-neutral regions.

### (c) Current-Voltage Characteristic

The equilibrium minority-carrier concentrations are

$$p_{n0} = \frac{n_i^2}{N_D}, \quad n_{p0} = \frac{n_i^2}{N_A}$$

The calculated minority-carrier concentrations are

$$p_{n0} = 1.0 \times 10^4 \text{ cm}^{-3}, n_{p0} = 1.0 \times 10^3 \text{ cm}^{-3}.$$

The reverse saturation current is

$$I_S = qA \left( \frac{D_p p_{n0}}{L_p} + \frac{D_n n_{p0}}{L_n} \right).$$

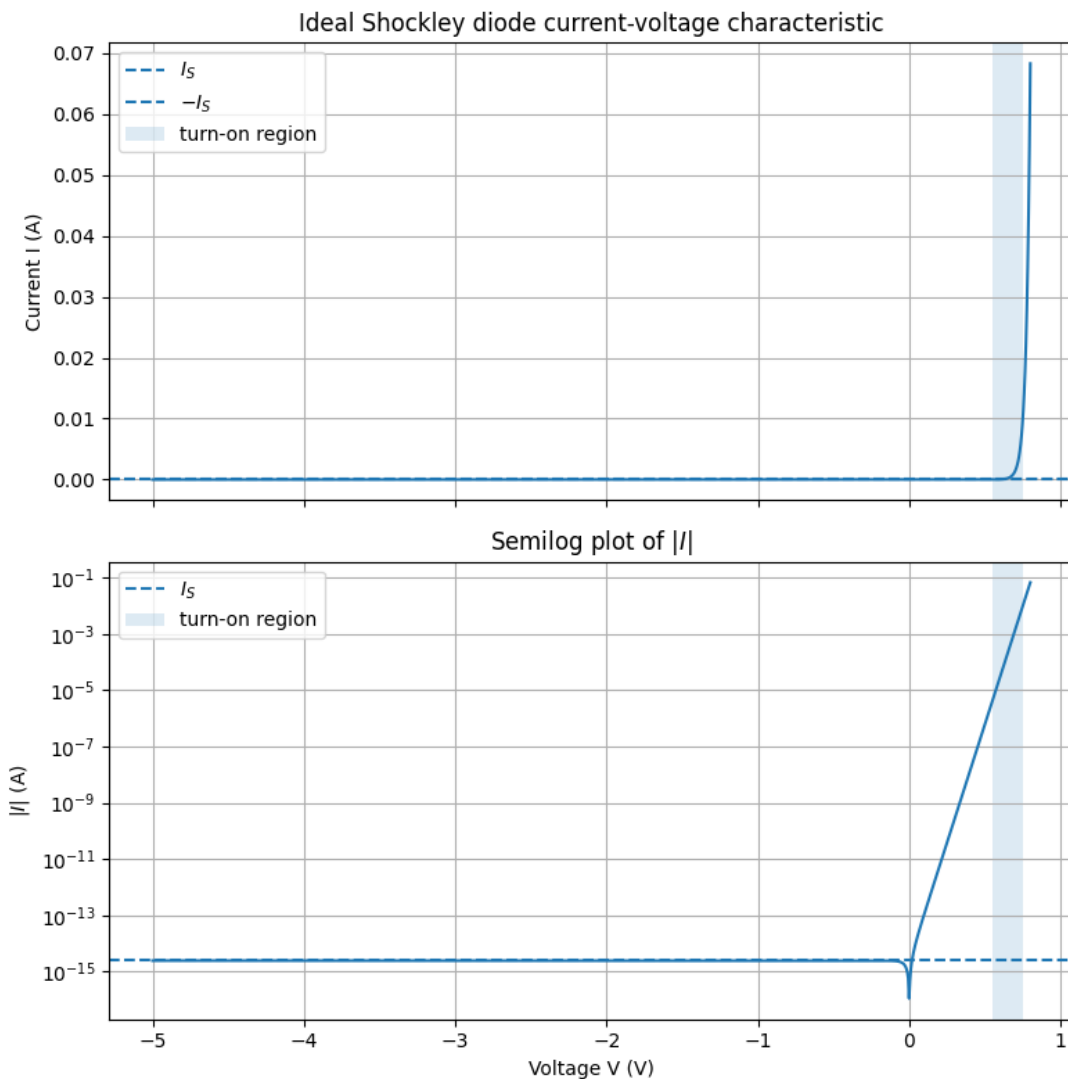
Using the assumed device area, diffusion coefficients, and diffusion lengths, the calculated saturation current is  $I_S = 2.483 \times 10^{-15} \text{ A}$ .

The ideal Shockley diode equation is

$$I(V) = I_S \left[ \exp \left( \frac{qV}{k_B T} \right) - 1 \right] I_S \left[ \exp \left( \frac{V}{V_T} \right) - 1 \right].$$

The current-voltage characteristic was simulated over the voltage range

$$-5 \text{ V} \leq V \leq 0.8 \text{ V}.$$



**Figure 3.** Ideal Shockley diode current-voltage characteristic shown using linear and semilogarithmic current scales.

In the forward-bias region, the diode current increases exponentially with voltage. A significant increase in current occurs around 0.6 to 0.7 V, which is identified as the turn-on region.

Under reverse bias, the ideal diode current approaches

$$I \approx -I_S.$$

Because  $I_S$  is extremely small, the reverse current appears approximately zero on the linear plot. However, the reverse saturation current can be clearly observed on the semilogarithmic plot.

The small downward spike near  $V=0$  in the semilogarithmic plot is not a physical effect. It occurs because

$$I(0) = 0,$$

and  $\log_{10}|I|$  is undefined when the current is exactly zero.

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#### **(d) Bias-Dependent Band Diagram**

When an external voltage ( $V$ ) is applied, the effective potential barrier becomes

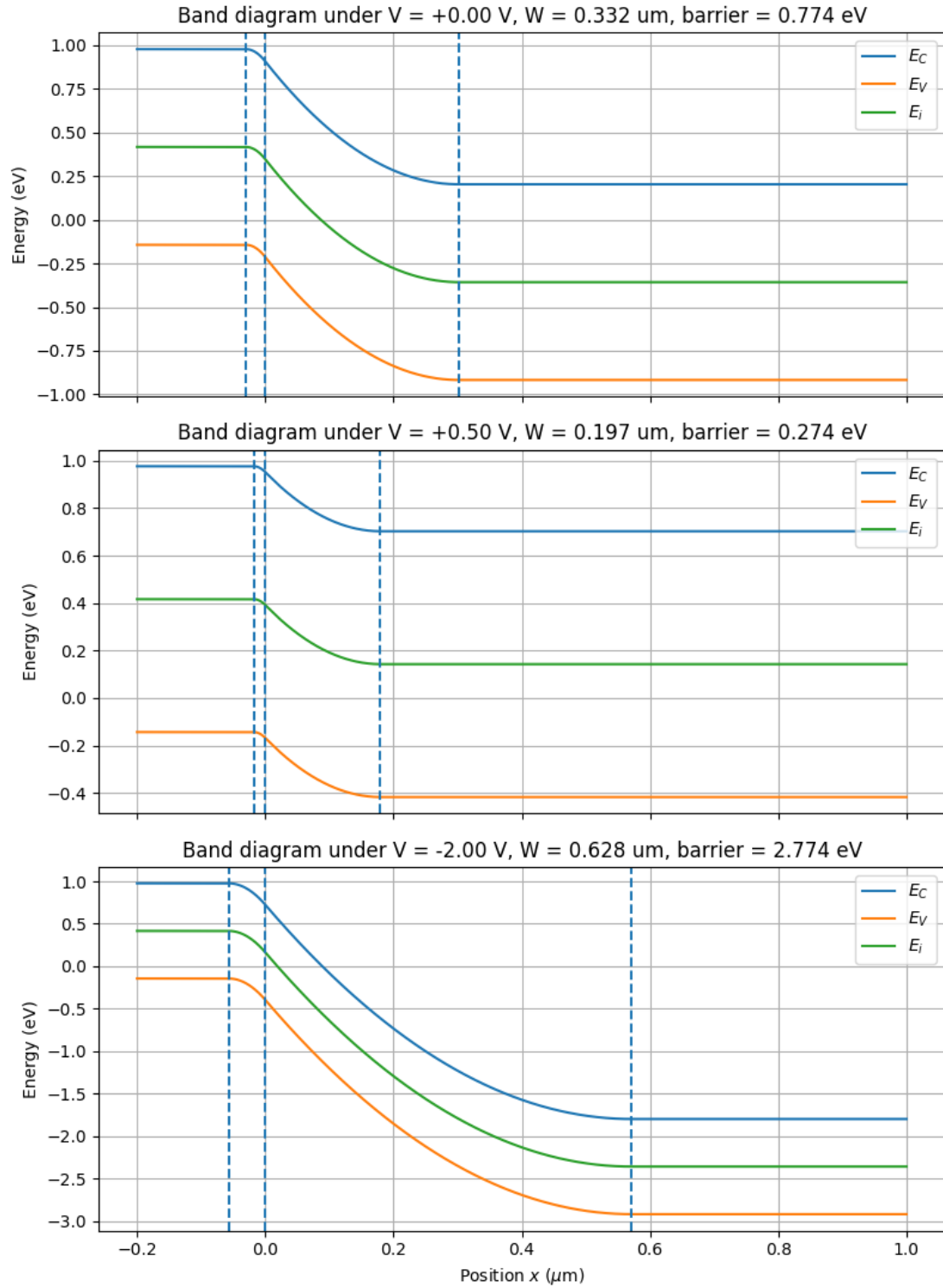
$$V_{\text{barrier}} = V_{bi} - V.$$

The bias-dependent depletion width is

$$W(V) = \sqrt{\frac{2\epsilon_s}{q} \left( \frac{1}{N_A} + \frac{1}{N_D} \right) (V_{bi} - V)}$$

Three bias conditions were simulated:

$$V = 0 \text{ V}, \quad V = +0.5 \text{ V}, \quad V = -2.0 \text{ V}.$$



**Figure 4.** Energy-band diagrams under equilibrium, forward bias, and reverse bias.

At equilibrium,



$$V = 0 \text{ V},$$

the potential barrier and depletion width are

$$V_{\text{barrier}} = 0.77384 \text{ V},$$

$$W = 0.33178 \text{ }\mu\text{m}.$$

For a forward bias of

$$V = +0.5 \text{ V},$$

the effective barrier is reduced to

$$V_{\text{barrier}} - 0.5 = 0.27384 \text{ V}$$

The depletion width decreases to  $W = 0.197 \text{ }\mu\text{m}$

The reduced barrier allows majority carriers to cross the junction more easily.

Therefore, the forward current increases exponentially, as observed in the current-voltage characteristic.

For a reverse bias of

$$V = -2.0 \text{ V}$$

the effective barrier becomes

$$V_{\text{barrier}} = V_{bi} - (-2.0) = 2.77384 \text{ V}$$

The depletion width increases to  $W = 0.628 \text{ }\mu\text{m}$ .

The larger barrier prevents majority carriers from crossing the junction. Therefore, before breakdown occurs, the reverse current remains close to the small reverse saturation current.

The simulated depletion widths and barrier heights are summarized below.

| Applied voltage | Bias condition | Barrier height | Depletion width     |
|-----------------|----------------|----------------|---------------------|
| 0 V             | Equilibrium    | 0.774 eV       | 0.332 $\mu\text{m}$ |
| +0.5 V          | Forward bias   | 0.274 eV       | 0.197 $\mu\text{m}$ |
| −2.0 V          | Reverse bias   | 2.774 eV       | 0.628 $\mu\text{m}$ |

This simplified bias-dependent band model modifies the equilibrium band bending by replacing  $V_{bi}$  with  $V_{bi} - V$ . The electron and hole quasi-Fermi levels were not explicitly solved in the present depletion-approximation model.

The ideal Shockley diode model neglects several physical mechanisms that exist in a real p-n junction. In this section, recombination-generation current, series resistance, and reverse breakdown are added to investigate how they modify the ideal current-voltage characteristic.

## 2.1 Shockley-Read-Hall Recombination-Generation Current

The ideal Shockley diode equation considers minority-carrier diffusion in the quasi-neutral p-type and n-type regions. However, electron-hole recombination and generation can also occur through defect states inside the depletion region. To investigate this effect, a simplified Shockley-Read-Hall recombination-generation current was added to the ideal diode model.

### 2.1.1 Model and Assumptions

The following assumptions were used:

1. The dominant defect level is located near the middle of the silicon band gap.
2. The electron and hole lifetimes are equal.
3. The carrier lifetime is uniform throughout the depletion region.
4. The depletion approximation remains valid.
5. The applied forward voltage remains below the built-in voltage.

The Shockley-Read-Hall carrier lifetime was assumed to be

$$1.0 \times 10^{-6} \text{ s}$$

The bias-dependent depletion width is

$$W(V) = \sqrt{\frac{2\epsilon_s}{q} \left( \frac{1}{N_A} + \frac{1}{N_D} \right) (V_{bi} - V)}$$

The characteristic recombination-generation current scale was calculated as

$$\frac{qAn_iW(V)}{2\tau_{SRH}}$$

A simplified continuous expression was used for the depletion-region recombination-generation current:

$$I_{RG0}(V) \left[ \exp\left(\frac{V}{2V_T}\right) - 1 \right]$$

The total diode current is therefore  $I_{diff} + I_{RG}$

where the ideal diffusion current is

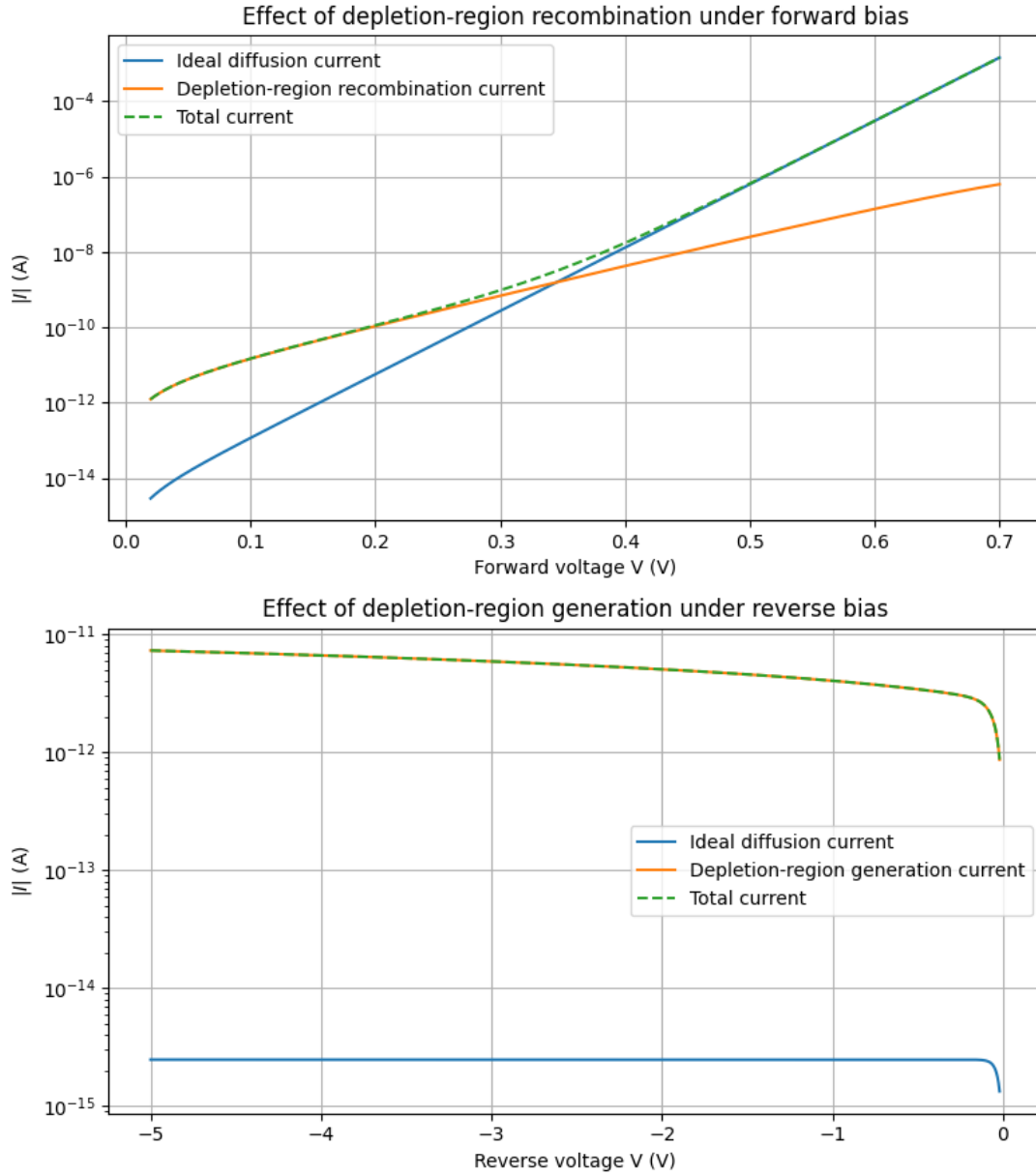
$$I_S \left[ \exp\left(\frac{V}{V_T}\right) - 1 \right]$$

The recombination current has an exponential dependence of approximately

$$I_{RG} \propto \exp\left(\frac{V}{2V_T}\right)$$

which corresponds to an ideality factor close to 2. In comparison, the ideal diffusion current has an ideality factor of 1.

### 2.1.2 Simulation Results



**Figure 5.** Comparison of the ideal diffusion current, depletion-region recombination-generation current, and total diode current under forward and reverse bias.

At equilibrium, the calculated recombination-generation current prefactor is

$$2.66 \times 10^{-12} \text{ A}$$

The ideal diffusion saturation current obtained previously is

$$2.483 \times 10^{-15} \text{ A}$$

Therefore,

$$\frac{I_{\text{RG0}}(0)}{I_s} \approx 1.07 \times 10^3$$

This indicates that depletion-region generation can produce a reverse leakage current much larger than the ideal diffusion saturation current for the selected carrier lifetime.

Under low forward bias, the recombination current dominates over the ideal diffusion current. For example, at

$$V = 0.2 \text{ V}$$

the calculated currents are approximately

$$I_{\text{diff}} = 5.69 \times 10^{-12} \text{ A}, I_{\text{RG}} = 1.07 \times 10^{-10} \text{ A}$$

Therefore, the recombination current is approximately nineteen times larger than the ideal diffusion current at this voltage.

As the forward voltage increases, the diffusion current eventually becomes dominant because it varies as  $\exp\left(\frac{V}{V_T}\right)$ , whereas the recombination current varies more slowly as  $\exp\left(\frac{V}{2V_T}\right)$

The two current components become approximately equal at

$$V \approx 0.345 \text{ V}$$

At  $V = 0.4 \text{ V}$  the ideal diffusion current is approximately

$$1.30 \times 10^{-8} \text{ A}$$

while the recombination current is approximately  $4.23 \times 10^{-9} \text{ A}$

Thus, the ideal diffusion current dominates at moderate and high forward bias.

Under reverse bias,

$$\exp\left(\frac{V}{2V_T}\right) \rightarrow 0$$

and the recombination-generation expression approaches

$I_{\text{RG}} \approx -I_{\text{RG0}}(V)$  This negative current represents electron-hole pair generation inside the depletion region. The generated carriers are separated by the junction electric field and contribute to the reverse leakage current.

The magnitude of the generation current increases gradually with reverse voltage

because the depletion width increases according to

$$W(V) \propto \sqrt{V_{bi} - V}$$

For example, at a reverse voltage of

$$V = -2 \text{ V}$$

the depletion width increases to approximately

$$W = 0.628 \text{ } \mu\text{m}$$

and the generation current is approximately

$$-5.03 \times 10^{-12} \text{ A}$$

At  $V = -5 \text{ V}$  the depletion width increases further to approximately  $W = 0.906 \text{ } \mu\text{m}$

and the generation current becomes approximately

$$-7.26 \times 10^{-12} \text{ A}$$

Therefore, unlike the constant reverse saturation current predicted by the ideal Shockley equation, the reverse leakage current increases gradually with reverse bias when depletion-region generation is included.

### 2.1.3 Discussion

The simulation shows that depletion-region recombination-generation mainly modifies the low-current regions of the diode current-voltage characteristic.

Under low forward bias, recombination through defect states produces an ideality factor close to 2 and causes the current to exceed the ideal diffusion-current prediction. At higher forward bias, the ideal diffusion current eventually becomes dominant because of its stronger exponential voltage dependence.

Under reverse bias, thermal generation of electron-hole pairs inside the depletion region produces a leakage current that is significantly larger than the ideal diffusion saturation current. The generation current also increases with reverse voltage because the depletion region becomes wider.

The present model is a simplified Shockley-Read-Hall approximation. It does not explicitly solve the spatial distributions of electron concentration, hole concentration,

trap occupation, or quasi-Fermi levels throughout the depletion region. Nevertheless, it reproduces the main qualitative effects of depletion-region recombination and generation on the diode current-voltage characteristic.

## 2.2 Series Resistance

The ideal diode model assumes that the p-type and n-type quasi-neutral regions are perfect conductors. In a real device, however, these semiconductor regions have finite resistivity. Therefore, part of the externally applied voltage is dropped across the bulk series resistance rather than across the p-n junction.

In the present model, an effective lumped series resistance of

$$R_s = 5.0 \, \Omega$$

was assumed. This resistance represents the combined resistance of the quasi-neutral p-type and n-type regions. Contact resistance was not modeled separately.

### 2.2.1 Series-Resistance Model

The externally applied terminal voltage is divided between the p-n junction and the series resistance:

$$V = V_D + IR_s$$

where  $V_D$  is the actual voltage across the p-n junction. Therefore,

$$V_D = V - IR_s$$

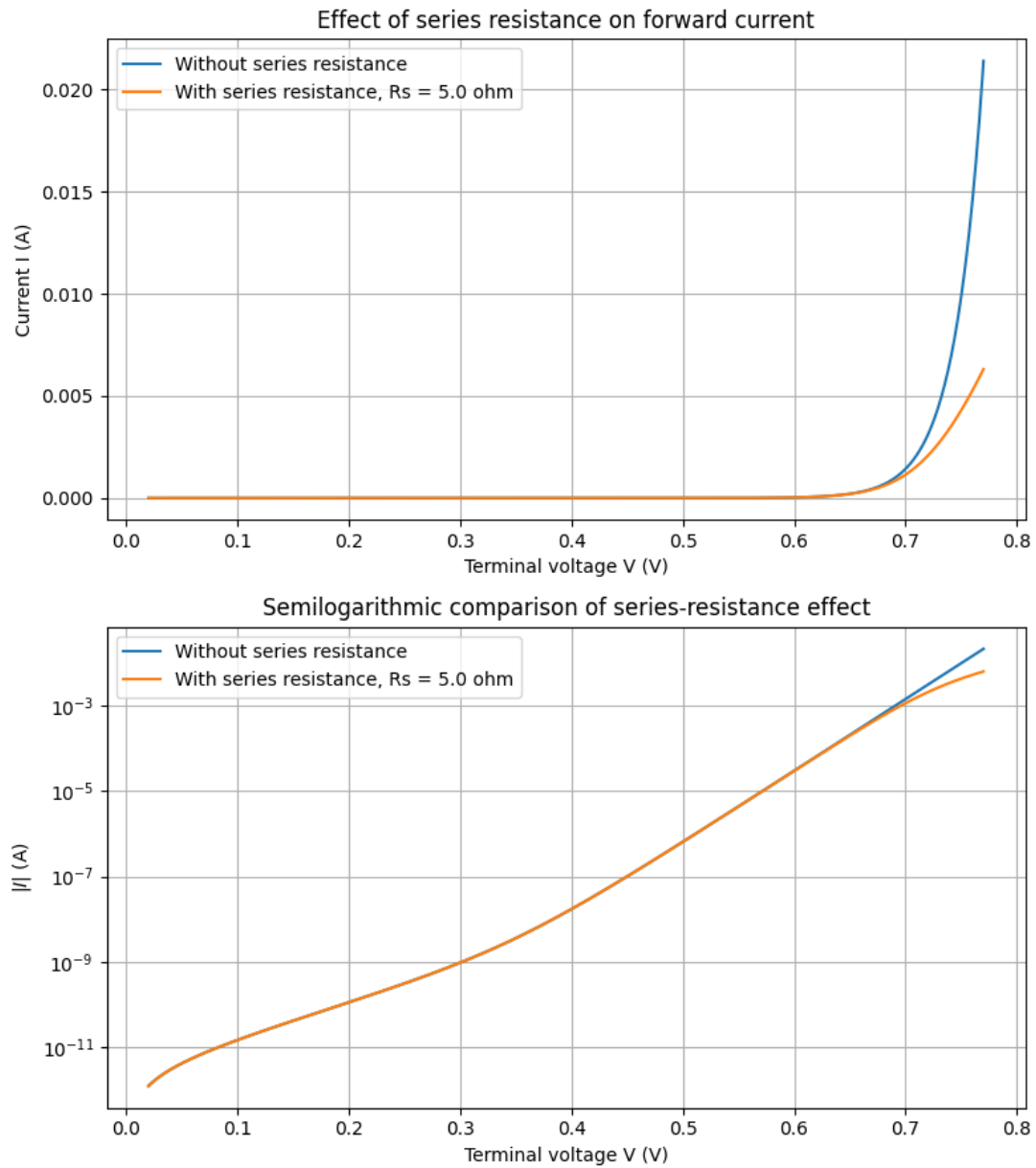
The diode current including the recombination-generation current is evaluated using the junction voltage:  $I = I_{\text{diff}} + I_{\text{RG}}$

Substituting  $V_D = V - IR_s$  gives

$$I = I_s \left[ \exp\left(\frac{V - IR_s}{V_T}\right) - 1 \right] + I_{\text{RG0}} \left[ \exp\left(\frac{V - IR_s}{2V_T}\right) - 1 \right]$$

Because the current  $I$  appears on both sides of this equation, the equation is implicit

and cannot be evaluated using a direct substitution. A bisection method was therefore used to solve the current numerically at each applied terminal voltage.



**Figure 6.** Forward current-voltage characteristics with and without an effective series resistance of ( $5\ \Omega$ ), shown using linear and semilogarithmic current scales.

### 2.2.2 Simulation Results

At low forward current, the voltage drop across the series resistance is negligible because

$$IR_s \ll V$$

Therefore, the current-voltage characteristics with and without series resistance are



nearly identical in the low-current region.

For example, at  $V = 0.60 \text{ V}$

the current without series resistance is approximately

$I_{\text{without } R_s} = 2.996 \times 10^{-5} \text{ A}$  whereas the current with series resistance is  $I_{\text{with } R_s} = 2.979 \times 10^{-5} \text{ A}$ .

The voltage drop across the series resistance is only

$IR_s = 1.49 \times 10^{-4} \text{ V}$  Therefore, series resistance has almost no effect at this voltage.

At a terminal voltage of

$$V = 0.70 \text{ V}$$

the current without series resistance is

$$I_{\text{without } R_s} = 1.428 \times 10^{-3} \text{ A}$$

while the current with series resistance decreases to

$$I_{\text{with } R_s} = 1.144 \times 10^{-3} \text{ A}$$

The voltage drop across the series resistance becomes

$$IR_s = 5.72 \times 10^{-3} \text{ V}$$

and the actual junction voltage is

$$V_D = 0.700 - 0.00572 \text{ V} = 0.69428 \text{ V}$$

The reduction in junction voltage decreases the forward current because the diode current depends exponentially on  $V_D$

At a higher terminal voltage of  $V = 0.75 \text{ V}$

the current predicted without series resistance is

$$I_{\text{without } R_s} = 9.875 \times 10^{-3} \text{ A}$$

After including the  $5 \Omega$  series resistance, the current decreases to

$$I_{\text{with } R_s} = 4.300 \times 10^{-3} \text{ A}$$

The corresponding voltage drop is

$$IR_s = 2.15 \times 10^{-2} \text{ V}$$

so the actual voltage across the p-n junction is only

$$V_D = 0.750 - 0.0215 = 0.7285 \text{ V}$$

Thus, the current is reduced by approximately 56.5%.

### 2.2.3 Discussion

The simulation shows that series resistance mainly affects the high-forward-current region. At low current, the product  $IR_s$  is very small, and nearly all of the applied terminal voltage appears across the p-n junction. Therefore, the diode follows approximately exponential behavior.

As the forward current increases, the voltage drop  $IR_s$  becomes significant. The actual junction voltage is then smaller than the externally applied terminal voltage. Since the forward diode current depends exponentially on junction voltage, even a relatively small resistive voltage drop can produce a large reduction in current.

On the semilogarithmic current-voltage plot, the ideal exponential characteristic appears approximately as a straight line. After series resistance is included, the curve bends away from this straight-line behavior at high current. This bending represents the transition from junction-limited conduction to resistance-limited conduction.

When the series resistance becomes dominant, an additional increase in terminal voltage appears mainly across  $R_s$ , rather than across the junction. Therefore, the high-current characteristic becomes approximately linear instead of exponential.

The present model treats  $R_s$  as a lumped parameter. A more detailed model could calculate the p-side and n-side bulk resistances separately from their carrier mobilities, doping concentrations, cross-sectional area, and quasi-neutral-region lengths.

## 2.3 Reverse Breakdown

The ideal Shockley diode equation predicts that the reverse current approaches a small saturation current. However, when the reverse voltage becomes sufficiently large, a real p-n junction exhibits a rapid increase in reverse current. This phenomenon is known as reverse breakdown.

In this simulation, an empirical reverse-breakdown model was used. The assumed breakdown-voltage magnitude was

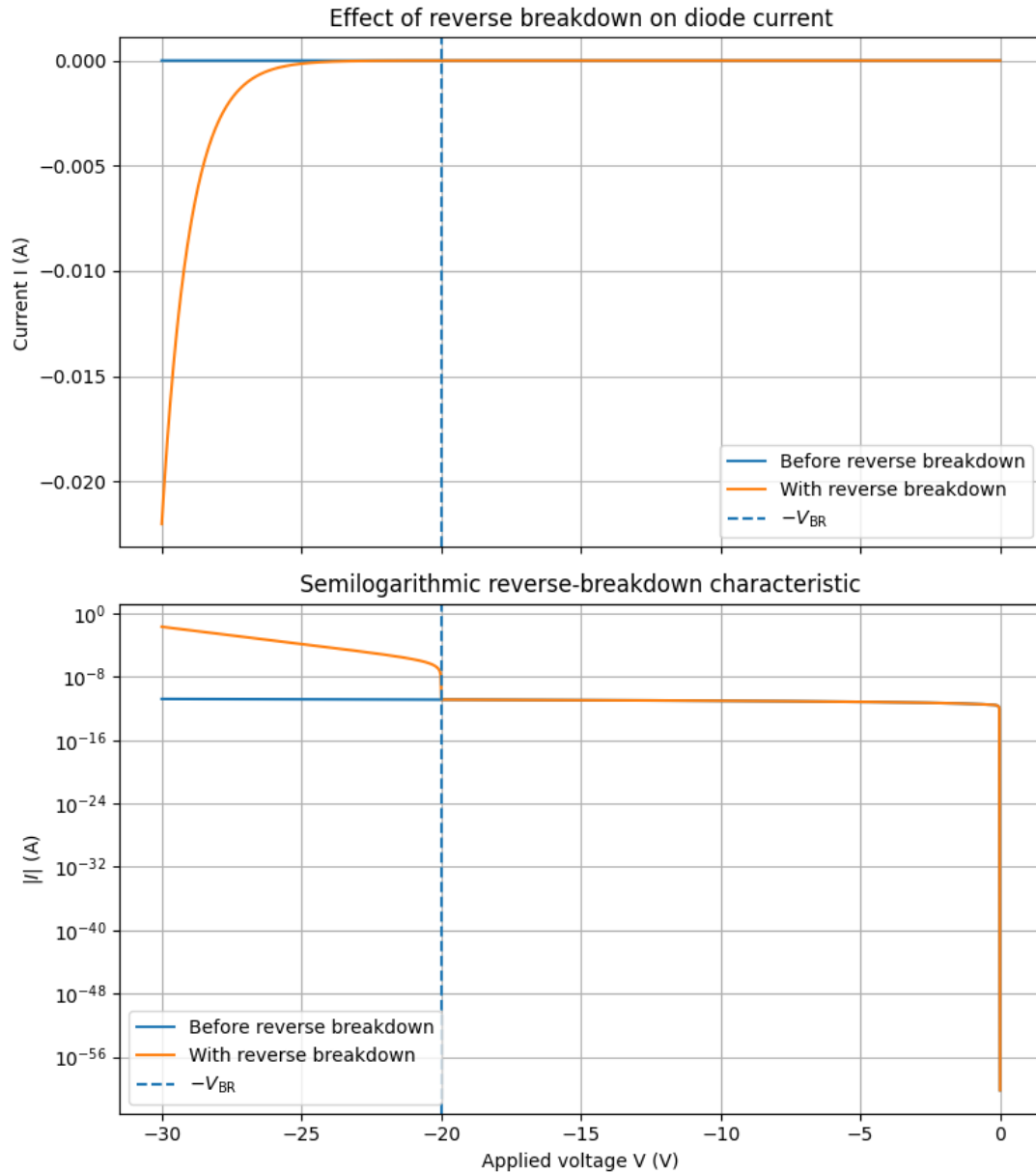
$$V_{\text{BR}} = 20 \text{ V}$$

The additional breakdown-current component was defined as

$$I_{\text{BD}}(V) = \begin{cases} 0, & V \geq -V_{\text{BR}} \\ I_{\text{BR}0} \left[ \exp\left(\frac{-V - V_{\text{BR}}}{V_A}\right) - 1 \right], & V < -V_{\text{BR}} \end{cases}$$

where  $I_{\text{BR}0} = 1.0 \times 10^{-6} \text{ A}$  and  $V_A = 1.0 \text{ V}$  the total current was calculated as

$$I_{\text{diode}} + I_{\text{BD}}$$



**Figure 7.** Reverse current-voltage characteristics before and after including the empirical reverse-breakdown model, shown using linear and semilogarithmic current scales.

Before the reverse voltage reaches the assumed breakdown voltage, the reverse current remains small and is mainly determined by the diffusion saturation current and depletion-region generation current.

When the applied voltage becomes more negative than  $-20$  V

the additional breakdown-current component increases rapidly and becomes the dominant current mechanism. The negative sign of the current indicates reverse

current flow.

This rapid increase may qualitatively represent avalanche multiplication or Zener tunneling. However, the present model is empirical and does not explicitly calculate impact-ionization coefficients or band-to-band tunneling probability. Therefore, the selected value of  $V_{BR} = 20$  V

is an assumed parameter rather than a breakdown voltage derived directly from the doping concentrations.

## 2.4 Metal-Semiconductor Contact

The previous diode models assumed ideal ohmic metal contacts. An ideal ohmic contact introduces no significant potential barrier and therefore does not limit the current delivered by the p-n junction. However, a real metal-semiconductor interface may form a Schottky barrier and introduce an additional voltage drop.

In the present model, one terminal was assumed to form a metal/n-type silicon Schottky contact, whereas the other terminal remained an ideal ohmic contact. The Schottky-contact current was described using the thermionic-emission equation

$$I_{MS} = I_{MS0} \left[ \exp \left( \frac{V_{MS}}{n_{MS} V_T} \right) - 1 \right]$$

The saturation-current parameter is

$$I_{MS0} = AA^{**} T^2 \exp \left( - \frac{\Phi_{Bn}}{V_T} \right),$$

where  $A^{**}$  is the effective Richardson constant and  $\Phi_{Bn}$  is the metal-semiconductor barrier height.

The parameters used in this model were

$$A^{**} = 112 \text{ A, cm}^{-2} \text{ K}^{-2}, \Phi_{Bn} = 0.55 \text{ eV}, n_{MS} = 1.10$$

The voltage drop across the Schottky contact can be written as

$$V_{MS} = n_{MS} V_T \ln \left( 1 + \frac{I}{I_{MS0}} \right)$$

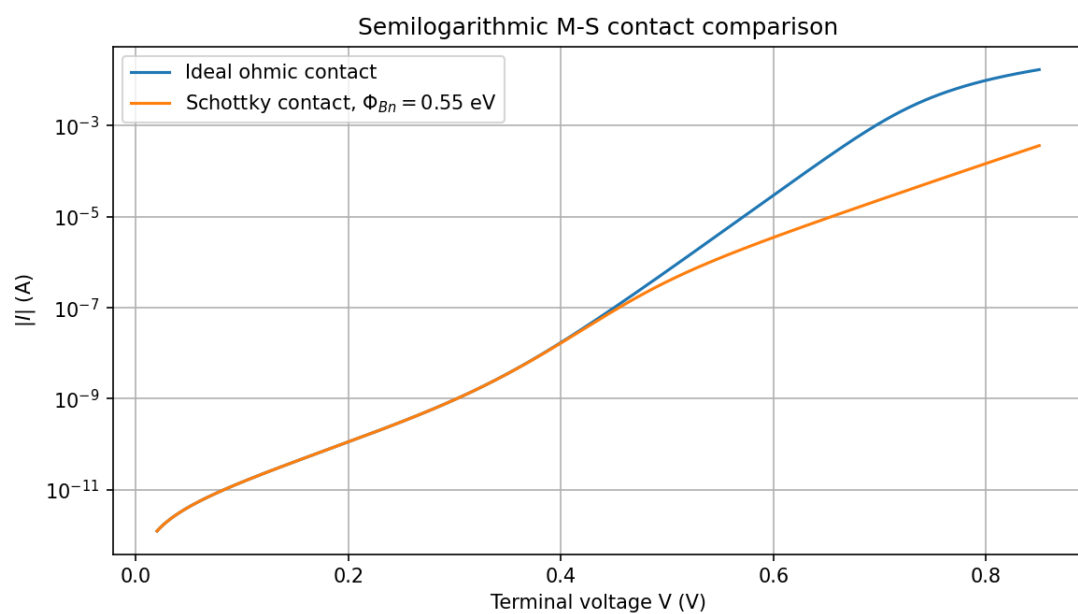
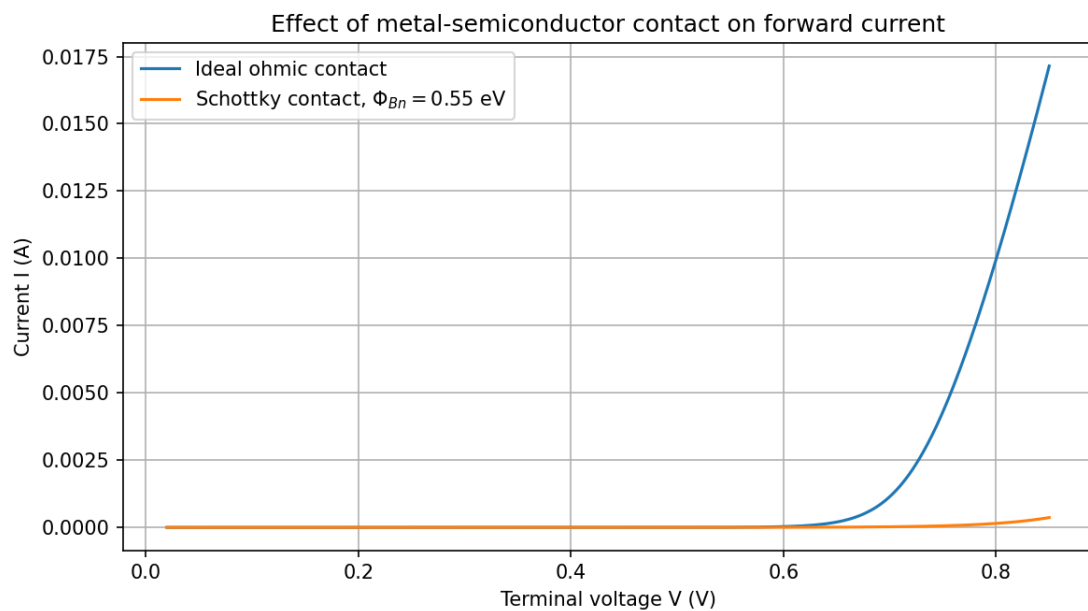
The terminal voltage is divided among the p-n junction, series resistance, and metal-semiconductor contact:

$$V = V_{PN} + IR_s + V_{MS}$$

Therefore, the actual p-n junction voltage is

$$V_{PN} = V - IR_s - V_{MS}$$

Because the current appears implicitly in both the series-resistance voltage drop and the metal-semiconductor contact voltage, the terminal current was solved numerically using the bisection method.



**Figure 8.** Forward current-voltage characteristics with an ideal ohmic contact and a non-ideal Schottky contact, shown using linear and semilogarithmic current scales.

The simulation shows that the Schottky contact reduces the terminal current relative to the ideal ohmic-contact case. As the current increases, part of the applied voltage is required to overcome the metal-semiconductor barrier. Consequently, the voltage available across the p-n junction is reduced.

The present model uses an assumed barrier height and a thermionic-emission approximation. It does not include image-force barrier lowering, tunneling through the contact, interface-state effects, or Fermi-level pinning. Therefore, the simulation is intended to demonstrate qualitatively how a non-ideal metal-semiconductor contact can limit the current delivered by a p-n junction.

結論：

本期末報告成功建立一個一維 abrupt silicon p-n junction diode 數值模型。利用 depletion approximation，計算得到 equilibrium built-in potential 為

$$V_{bi} = 0.03016 \mu\text{m}$$

p-side 與 n-side 的 depletion widths 分別為

$$x_p = 0.03016 \mu\text{m} \text{ 及 } x_n = 0.30162 \mu\text{m},$$

因此總 depletion width 為

$$W = 0.33178 \mu\text{m}.$$

由於 acceptor concentration 是 donor concentration 的十倍，空乏區主要延伸至摻雜濃度較低的 n-side。此結果符合 charge-neutrality condition：

$$N_A x_p = N_D x_n$$

模擬所得 charge density、electric field 與 electrostatic potential 均符合 abrupt p-n junction 的預期行為。Electric field 在 depletion region 外為零，並在 metallurgical junction 附近達到最大值。Electrostatic potential 則由 p-side 往 n-side 上升，其總變化量等於 built-in potential。

Equilibrium energy-band diagram 顯示能帶由 p-side 往 n-side 向下彎曲，總 band bending 約為

$$qV_{bi} = 0.774 \text{ eV.}$$

平衡狀態下 Fermi level 保持水平，表示元件內沒有淨電流流動。

理想 current–voltage characteristic 符合 Shockley diode equation。Forward bias 下，電流呈指數增加；reverse bias 下，電流則趨近於所計算的 saturation current：

$$I_S = 2.483 \times 10^{-15} \text{ A.}$$

Forward bias 會降低接面 barrier 與 depletion width，使載子更容易通過接面；reverse bias 則會提高 barrier 並增加 depletion width，因此抑制 majority-carrier current。

加入非理想效應後，不同機制會影響不同的操作區域。Shockley–Read–Hall recombination 會增加低 forward bias 下的電流，而 depletion-region generation 會增加 reverse leakage current。當 forward current 變大時，series resistance 造成明顯壓降，使電流偏離理想的指數特性。Empirical reverse-breakdown model 則成功呈現當外加電壓比假設的

$$V_{BR} = 20 \text{ V}$$

更負時，反向電流快速增加的現象。

Metal-semiconductor contact 模型顯示，Schottky barrier 會使 terminal current 明顯低於 ideal ohmic-contact case，因為部分外加電壓必須用於克服接觸能障，造成實際加在 p–n junction 上的電壓降低。

整體而言，理想 Shockley model 能夠描述 p–n junction 的基本行為，但若要模擬真實元件在較大電壓範圍內的特性，必須加入額外的物理機制。低電流區主要受到 recombination-generation current 影響，中等 forward bias 主要由 minority-carrier diffusion 主導，高 forward current 受到 series resistance 與 contact barrier 限制，而 large reverse bias 則由 breakdown current 主導。

目前模型仍有部分簡化。Electrostatics 使用 depletion approximation，而非 self-consistent Poisson solution；另外也未考慮 high-level injection、graded doping、band-gap narrowing、interface states、metal-semiconductor tunneling、image-force barrier lowering，以及由實際摻雜條件推導出的 avalanche 或 Zener breakdown voltage。



未來可加入上述效應，以建立更完整且更接近真實半導體元件的數值模型。